



Welcome to A Catholic Introduction to Artificial Intelligence Class 1! Course Overview & What is AI?

"The question is not whether intelligent machines can have any emotions, but whether machines can be intelligent without any emotions." – Marvin Minsky

1. Learning Objectives

By the end of this class, students will be able to:

1. Understand the course objectives, requirements, and expectations.
2. Define Artificial Intelligence (AI).
3. Distinguish between AI, Machine Learning (ML), and Deep Learning (DL).
4. Identify examples of AI in their daily lives.
5. Begin to think critically about what AI is and what it is not.

1A. Course Objectives

The Catholic Church views science to explore and comprehend the natural world created by God. Technology is encouraged when it respects ethical principles and upholds the dignity of human life. Catholics recognize the benefits of technological advancements for improving human life and society. Moral discernment is crucial to guide technology to serve humanity ethically and avoid harm or injustice.

This course integrates faith and technology to explore faith, ethics, and AI advancements together. This course will not teach students how to exploit AI concepts that violate academic integrity. Its main goal is to introduce the student AI concepts and the technology that makes it happen. Over the course of the topics, students will understand the so-called truth that AI provides and the truth that it doesn't provide. And it will cut through the fantasy and hype so often attributed to AI.

To summarize:

- Become critical thinkers and responsible users.
- Understand how AI impacts privacy, ethics, employment, and truth.

1B. Course Prerequisites

One of the following or **equivalent**:

- Introduction to Computer Science, Part 1 or Part 2
- Introduction to Computer Science Using SNAP!
- Computer Programming 101 with Python
- C# Programming with Unity 101

1C. Course Requirements

This course uses the following AI-based tools:

- **NotebookLM** is an AI research and study tool developed by Google that helps users understand and work with their own documents. Instead of answering questions from the general internet, it analyzes sources you upload—such as PDFs, Google Docs, or notes—and generates summaries, explanations, study guides, and even audio-style discussions based only on those materials. This makes it especially useful for research, learning, and organizing information because the AI stays grounded in the sources you provide. **A Google account is required.**
- **Jupyter Notebook** is an open-source interactive computing environment where users can write code, display results, add explanations, and create visualizations all in one document. It is widely used in data science, machine learning, and AI because it allows programmers to run code step-by-step while documenting their work. This makes it especially useful for learning, experimentation, and sharing reproducible research.

Jupyter Notebooks are best created and executed using either Visual Studio Code (software download required, no account required) or Google Colab (Google account required). Refer to the description of these platforms below (choose either VSC or Colab).

- **Visual Studio Code (VSC)** : VSC is an integrated development environment (IDE) that aids in the writing, debugging, and execution of a large range of computer programs. It also can handle Jupyter Notebooks. Students with computer programming experience and who are familiar with VSC will find using VSC for Jupyter Notebooks quite easy. VSC software is free but it must be installed on the student's computer or laptop.
- **Google Colab** : (short for Google Colaboratory) is a free cloud-based environment from Google that lets users write and run Python code in a web browser using Jupyter Notebooks. Because it runs in the cloud and can provide access to GPUs, students can build and test AI models without installing software on their own computers. **A Google account is required.**
- **Jupyter Lite** : Jupyter Lite is part of "Project Jupyter", a large umbrella project that covers many different software offerings and tools that includes Jupyter Notebook and JupyterLab, which are both popular notebook-editor programs. You access Jupyter Lite by going to <https://jupyter.org/try-jupyter/tree>.

Access to the following open-source Large Language Models (LLMs):

- **Ollama** (<https://ollama.com/>): Ollama (OLLAMA) is an open-source, large-scale language model developed by Alibaba Cloud. It can generate natural-sounding text responses based on the prompt provided, making it capable of understanding and engaging with complex topics across multiple domains. An Ollama free-usage account is required.

Access to one of the following cloud-based proprietary AI models:

- **ChatGPT** (<https://chatgpt.com/>): an AI assistant that explains ideas, answers questions, researches information, and helps with writing, learning, planning, and problem-solving across many topics (from Chat GPT). No account registration required.
- **Google Gemini** (<https://gemini.google.com/app>): I am Gemini 3 Flash, an adaptive AI collaborator. I'm here to help you create text, images, music, and video with clarity and wit (from Gemini). A Google account is required.
- **Claude** (<https://claude.ai/login>): an AI assistant made by Anthropic. I can answer questions, write, analyze, code, and help with a wide range of tasks (from Claude). Account registration is required.

1D. Course Expectations

This is a **live** course - you are expected to attend the live session each week. The recording is not intended to be viewed for the first time; it is intended as a refresher after you have attended the weekly live class.

You are expected to complete all assignments on time. There is a 20% late penalty for late assignments. Any assignment not submitted within two weeks of its due date will not be accepted.

This course will use the following Academic Integrity Policy (AIP) as it relates to the use of artificial intelligence. All other common-sense integrity policies also remain in effect.

1	NO AI	The assessment is completed entirely without AI assistance. This level ensures that students rely solely on their knowledge, understanding, and skills. AI must not be used at any point during the {class assignment.}
2	AI-ASSISTED IDEA GENERATION AND STRUCTURING	AI can be used in the assessment for brainstorming, creating structures, and generating ideas for improving work. No AI content is allowed in the AI must not be used at any point during the {class assignment.}.
3	AI-ASSISTED EDITING	AI can be used to make improvements to the clarity or quality of student created work to improve the final output, but no new content can be created using AI. AI can be used, but your original work with no AI content must be provided in an appendix.
4	AI TASK COMPLETION, HUMAN EVALUATION	AI is used to complete certain elements of the task, with students providing discussion or commentary on the AI-generated content. This level requires critical engagement with AI generated content and evaluating its output. You will use AI to complete specified tasks in your {assignment assessment.} Any AI created content must be cited.
5	FULL AI	AI should be used as a 'co-pilot' in order to meet the requirements of the assessment, allowing for a collaborative approach with AI and enhancing creativity. You may use AI throughout your {assignment assessment.} to support your own work and do not have to specify which content is AI generated.

1E. Assessment Overview

- Short essays - 30%
- Quizzes- 10%
- Labs - 20%
- Final Project - 40%

2. Define Artificial Intelligence (AI)

What is AI to you? Define AI in your own words.

When scientists speak of artificial intelligence, they are referring to the ways that a computer can be **intelligent** like humans - literally. It has little or nothing to do with human intelligence. Instead, scientists ask: Can a computer display **intelligence** ? How would a computer demonstrate that it is **intelligent** ? What would it have to do?

How do humans demonstrate **intelligence** ?

- **Learning:** Obtain, process, and maintain new information
- **Reasoning:** Manipulate and judge information
- **Understanding:** Analyze results of new information
- **Grasp truth:** Validate processed information
- **See relationships:** How data interacts with other data
- **Consider meaning:** Apply truths to situations
- **Separate facts from belief:** Determine if the data is adequately and consistently supports conclusions

AI Scientists categorize AI in four ways:

- Acting humanly

When a user can't differentiate between a computer and a human (Turing test)

- Thinking humanly

When a computer performs a task like a human (self-driving cars)

- Thinking rationally

When a computer solves problems logically like a human (checkers, chess)

- Acting rationally

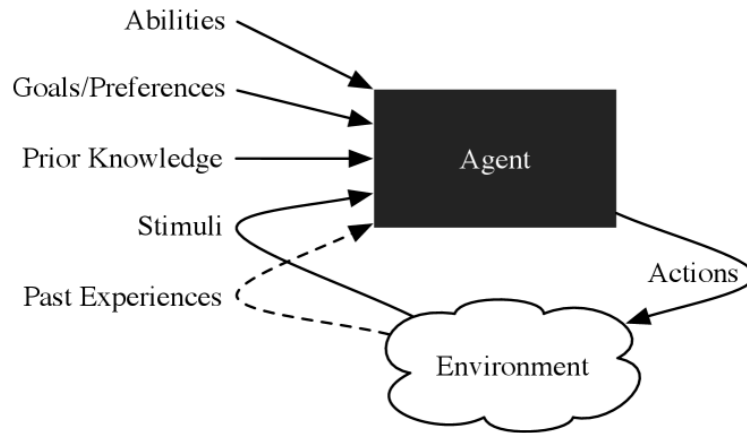
When a computer interacts with its environment to make a decision in real-time (logically deviating from normal processing when required)

This all leads to the main and crucial part of AI:

AI is about building agents that perceive and act to achieve goals. Agents perceive its environment through sensors and act using actuators.

Examples:

- thermostat
- self-driving car
- ChatGPT



An agent interacting with an environment (<https://artint.info/3e/html/ArtInt3e.Ch1.S3.html>)

2A: AI Capabilities

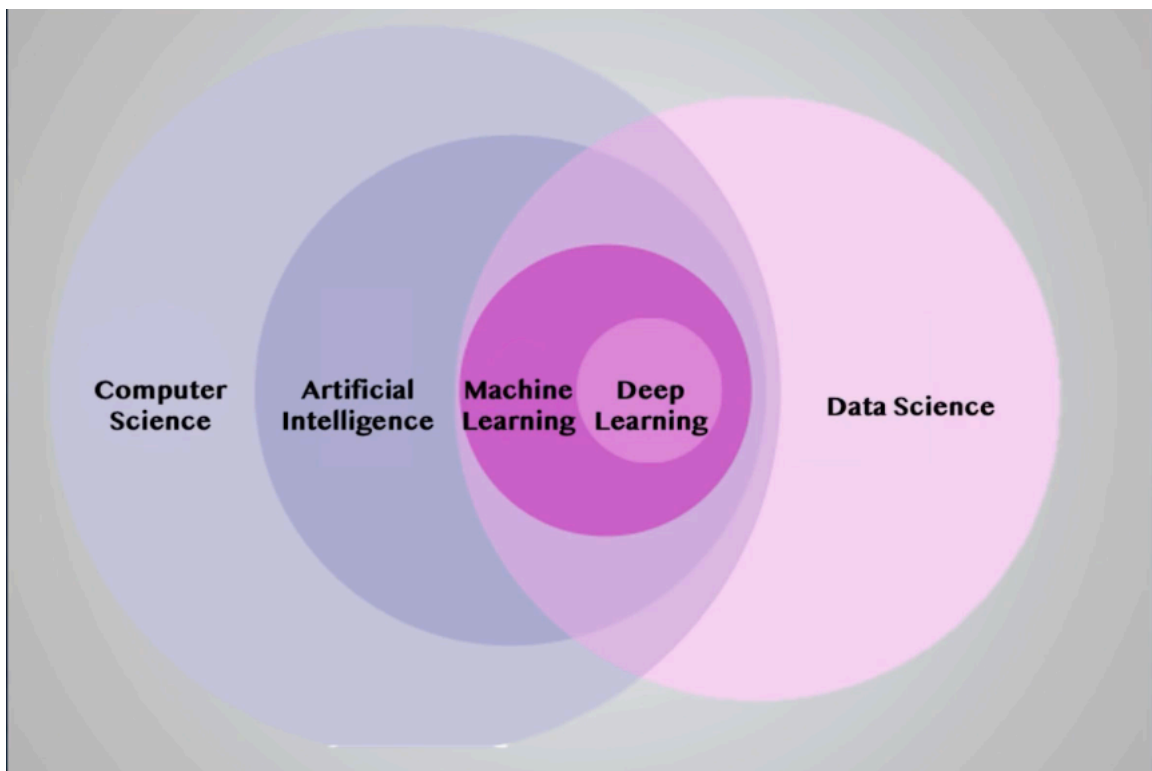
- **Narrow** AI that is good at one specific task (or a narrow set of tasks)
 - Examples are: spam filters, fraud detection models, recommendation systems, voice assistants, image classifiers, generative AI (ChatGPT, Copilot, Claude, etc.)
 - Each of these systems can only perform well in its own domain
 - Most AI systems today are **narrow** or **weak** AI
- **General** AI can learn and perform any intellectual tasks a human can
 - Not a pre-programmed task
 - Examples are: learn a new job without training, reason across domains, transfer knowledge
- **Superintelligence** AI that is more capable than humans at essentially all intellectual tasks
 - Can reason, create, discover, strategize, and socially understand all on its own
 - Far beyond humans

3. Distinguish between Computer Science, AI, Machine Learning (ML), and Deep Learning (DL).

AI is the broad concept of machines doing intelligent things.

- Machine Learning (ML) is a subset of AI that uses data and algorithms to "learn" from experience using labels.
- Deep Learning (DL) is a further subset of ML that uses artificial neural networks to model complex patterns.
- Analogy: Think of AI as the universe, ML as a galaxy, and DL as a solar system within that galaxy.
- Supervised Learning is a type of machine learning where a model is trained using **labeled data** — that is, data where the correct answers are already known. In this approach, the model learns by comparing its predictions to the correct answers and gradually improving over time. For example, if you train a model with many emails labeled "spam" or "not spam," it learns patterns that help it classify new emails correctly. *You show the computer examples with the right answers, and it learns to predict those answers on its own.*
- Unsupervised Learning is a type of machine learning where a model is given **unlabeled data** and must find patterns, groupings, or structure on its own. Unlike supervised learning, no one tells the model the "right answer" ahead of time. Instead, it looks for similarities or hidden relationships in the data. For example, a model might group customers into different categories based on buying behavior without being told what those categories should be. *The computer is given a pile of data and asked to discover what patterns are there by itself.*

Below is another way of describing this.



This diagram shows how several important technology fields are related to one another. The biggest circle is **Computer Science**, which is the broad field that includes programming, software, hardware, networks, databases, cybersecurity, and much more. Inside computer science is **Artificial Intelligence**, which focuses specifically on building systems that can do things that seem intelligent, such as recognizing speech, making decisions, or understanding language.

Inside artificial intelligence is **Machine Learning**, which is a more specific area where computers learn patterns from data instead of being told every rule directly by a programmer. For example, instead of programming every rule for identifying spam email, a machine learning system studies many examples of spam and non-spam messages and learns the patterns for itself.

Inside machine learning is **Deep Learning**, which is an even more specialized approach that uses layers of artificial “neurons” called neural networks. Deep learning is what powers many of today’s most advanced AI systems, such as image recognition, voice assistants, and large language models like ChatGPT.

The **Data Science** circle overlaps with AI and machine learning because these fields often use the same tools and data. However, data science is not exactly the same as AI. Data science is more focused on collecting, organizing, analyzing, and drawing conclusions from data. A data scientist might use machine learning, but they might also just use statistics, charts, and spreadsheets to answer questions.

The main idea students should understand is this: **AI is part of computer science, machine learning is part of AI, and deep learning is part of machine learning.** Data science overlaps with all of them because data is essential to how modern AI works.

4. Identify examples of AI in their daily lives.

Examples of AI in Everyday Life:

- Smartphones: Voice assistants, auto-correct.
- Streaming Services: Recommendation engines.
- Social Media: Feed algorithms, facial tagging.
- Navigation: Google Maps predicting traffic.
- Retail: Chatbots, personalized ads.

5. Ethics and Bias Teasers

- AI can make mistakes (e.g., false arrests due to facial recognition errors).
- Data used to train AI can be biased.
- Discussion: Should we always trust AI?
- Discussion: Who is responsible when AI is wrong?

6. Exit Ticket / Assessment:

"What is one way you've used AI without realizing it? What do you hope to learn in this course?"

7. Vocabulary

- **Artificial Intelligence (AI)** The field of computer science that focuses on creating systems capable of performing tasks that typically require human intelligence, such as problem-solving, understanding language, recognizing patterns, and making decisions.
- **Computer Science** The broad field of study that includes programming, computing systems, software, and technologies such as artificial intelligence.
- **Deep Learning (DL)** A subset of machine learning that uses artificial neural networks to model complex patterns.
- **Generative AI** AI systems that can create new content such as text, images, music, or other media.
- **Machine Learning (ML)** A subset of AI that uses data and algorithms to learn from experience.
- **Narrow AI** AI that is designed to perform one specific task or a narrow set of tasks.
- **Neural Networks** A type of computing model used in deep learning that is inspired by the structure of the human brain.
- **Superintelligence AI** A hypothetical AI that would be more capable than humans at essentially all intellectual tasks.
- **Supervised Learning** A type of machine learning that learns from data using labels.
- **Unsupervised Learning** A type of machine learning in which the system looks for patterns in data without being given labels.